We run a quick test to verify that our RAG model works as expected.

```
question = "What is GPT ?"
dialog = {"query": question}
rag_model.invoke({"query":"What is GPT? "})["result"]
```

```
'\nGenerative Pre-trained Transformers (GPTs) are a type of artificial intelligence model that
uses machine learning to generate human-like text based on input data. They are a subset of a
broader category of technologies known as general-purpose technologies (GPTs), which have the
potential to significantly impact a wide range of industries and applications. GPTs are capabl
e of performing a variety of tasks, including text generation, translation, summarization, and
question answering. They are trained on large amounts of text data and can generate coherent a
nd contextually relevant responses to a wide range of prompts. GPTs have the potential to sign
ificantly impact the labor market by automating certain tasks and potentially creating new typ
es of work.'
```

Figure 7.47 – Verifying our loaded model works as expected

Next, we use our RAG chatbot to generate answers for all of the example questions we stored in the `evaluation_table`. These responses are what we will compare against the ground truth answers. We build a `pandas_udf` to make this part run faster.

```
import mlflow
from mlia_utils.mlflow_funcs import *
from pyspark.sql.functions import col, udf, length, pandas_udf, explode

os.environ['DATABRICKS_TOKEN'] = dbutils.secrets.get("mlaction", "rag_sp_token")
model_name = f"{catalog}.{database_name}.mlaction_chatbot_model"

model_version_to_evaluate = get_latest_model_version(model_name)
mlflow.set_registry_uri("databricks-uc")
rag_model = mlflow.langchain.load_model(f"models:/{model_name}/
{model_version_to_evaluate}")

@pandas_udf("string")
def predict_answer(questions):
    def answer_question(question):
        dialog = {"query": question}
        return rag_model.invoke(dialog)['result']
    return questions.apply(answer_question)
```

Figure 7.48 – Creating a function to receive a question and return an answer using our RAG model